14 measured voices, 3 prompts, 2 runs per prompt

TTS LATENCY BENCHMARK ACROSS MODEL FAMILIES

We expanded the benchmark to cover the TTS model families currently exposed through the live Telnyx Voices API. The main number is first audio latency: time from sending text to receiving the first audio bytes a live voice agent can start playing.

FASTEST FIRST AUDIO P50

250 ms

Amazon Polly Standard, Joanna had the lowest first audio p50 in this run. Every measured voice completed 6/6 runs.

WEBSOCKET
Polly Standard, Joanna
250 ms
first audio p50

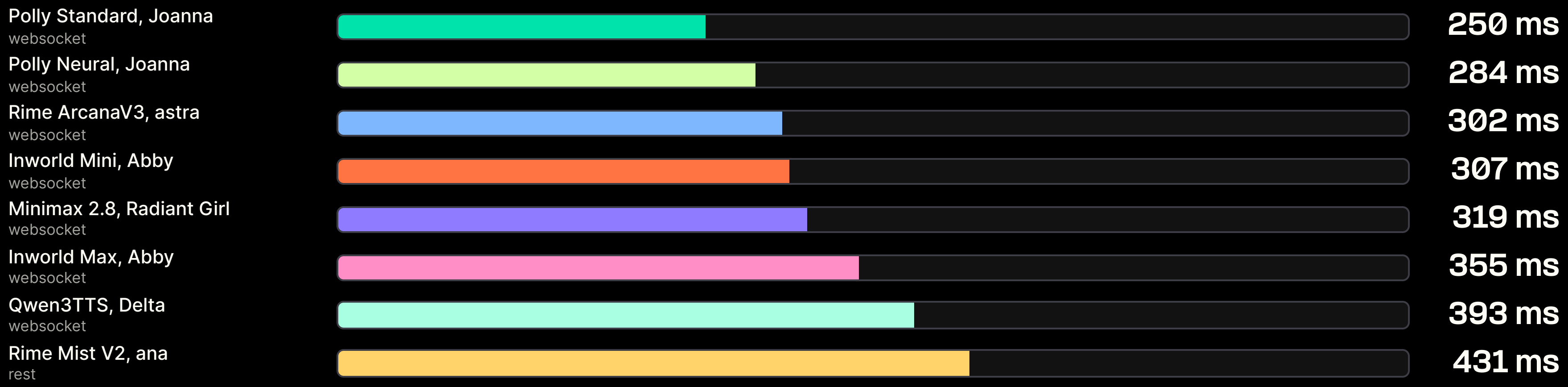
WEBSOCKET
Polly Neural, Joanna
284 ms
first audio p50

WEBSOCKET
Rime ArcanaV3, astra
302 ms
first audio p50

WEBSOCKET
Inworld Mini, Abby
307 ms
first audio p50

WEBSOCKET
Minimax 2.8, Radiant Girl
319 ms
first audio p50

WEBSOCKET
Inworld Max, Abby
355 ms
first audio p50



Prompt set: greeting, routing, and short agent response

Benchmark run, May 20, 2026

COVERAGE MAP

This page maps the model-family names to the representative voice IDs used in the benchmark. Ultra and Google were not included in the measured set.

REQUESTED MODEL FAMILY	REPRESENTATIVE TESTED	STATUS	PATH
Minimax 2.8 Turbo	Minimax 2.8 Turbo, Radiant Girl	Measured	WebSocket
Rime ArcanaV3	Rime ArcanaV3, astra	Measured	WebSocket
Rime Mist V2	Rime Mist V2 via Natural, ana	Measured	REST
Inworld Mini	Inworld Mini, Abby	Measured	WebSocket
Inworld Max	Inworld Max, Abby	Measured	WebSocket
Qwen3TTS	Qwen3TTS, Delta	Measured	WebSocket
xAI TTS	xAI TTS, Eve	Measured	WebSocket
Resemble	Resemble Turbo and Pro	Measured	REST
Amazon Polly Standard	Amazon Polly Standard, Joanna	Measured	WebSocket
Amazon Polly Neural	Amazon Polly Neural, Joanna	Measured	WebSocket
Azure Neural	Azure Neural, Jenny	Measured	WebSocket
Azure Multilingual Neural	Azure Multilingual Neural, Ava	Measured	WebSocket
Azure Dragon HD	Azure Dragon HD, Ava	Measured	WebSocket
Telnyx Ultra	Not included	Not measured	Not accessible in this pass
Google Neural	No voice ID in live API	Not measured	Not exposed by Voices API
Google Chirp 3 HD	No voice ID in live API	Not measured	Not exposed by Voices API
Azure Standard	No standard voice in live API	Not measured	Only Neural and Dragon HD surfaced

Source: live Telnyx Voices API snapshot, 3,295 voices

Representative English voices were selected where possible.

WHAT THE RUN SAYS

All measured voices returned successful audio for 6/6 runs. Amazon Polly Standard and Neural had the fastest first audio p50 in this run, followed by Rime ArcanaV3, Inworld Mini, and Minimax 2.8 Turbo.

FASTEST START

Polly Standard leads first audio

Amazon Polly Standard, Joanna finished at 250 ms first audio p50.

COVERAGE CORRECTION

Natural maps to Rime Mist V2

Rime Mist V2 is represented by Telnyx Natural, ana over REST.

GOOGLE CAVEAT

Google was not exposed

No Google Neural or Chirp 3 HD voice IDs were present in the live Voices API.

MODEL FAMILY AND VOICE	INTERFACE	FIRST AUDIO P50	FIRST AUDIO P95	FINAL AUDIO P50	SUCCESS
Amazon Polly Standard, Joanna	websocket	250 ms	400 ms	250 ms	6/6
Amazon Polly Neural, Joanna	websocket	284 ms	314 ms	284 ms	6/6
Rime ArcanaV3, astra	websocket	302 ms	563 ms	1722 ms	6/6
Inworld Mini, Abby	websocket	307 ms	367 ms	751 ms	6/6
Minimax 2.8 Turbo, Radiant Girl	websocket	319 ms	362 ms	565 ms	6/6
Inworld Max, Abby	websocket	355 ms	440 ms	1133 ms	6/6
Qwen3TTS, Delta	websocket	393 ms	552 ms	1765 ms	6/6
Rime Mist V2 via Natural, ana	rest	431 ms	497 ms	538 ms	6/6
Azure Neural, Jenny	websocket	462 ms	578 ms	503 ms	6/6
Azure Multilingual Neural, Ava	websocket	471 ms	722 ms	557 ms	6/6
Resemble Pro, Aaron	rest	595 ms	737 ms	1234 ms	6/6
Azure Dragon HD, Ava	websocket	683 ms	702 ms	1778 ms	6/6
xAI TTS, Eve	websocket	726 ms	838 ms	1548 ms	6/6
Resemble Turbo, Alex	rest	729 ms	880 ms	1293 ms	6/6

HOW TO READ THIS BENCHMARK

The benchmark sends the same prompt set to each representative voice and records when audio first arrives and when the final audio bytes arrive.

MAIN METRIC
First audio p50
Median time from sending text to receiving first audio bytes.

COMPLETION TIMING
Final audio p50
Median time from sending text to last observed audio bytes.

TRANSPORT
Interface
WebSocket for most providers. REST for Natural and Resemble in this run.

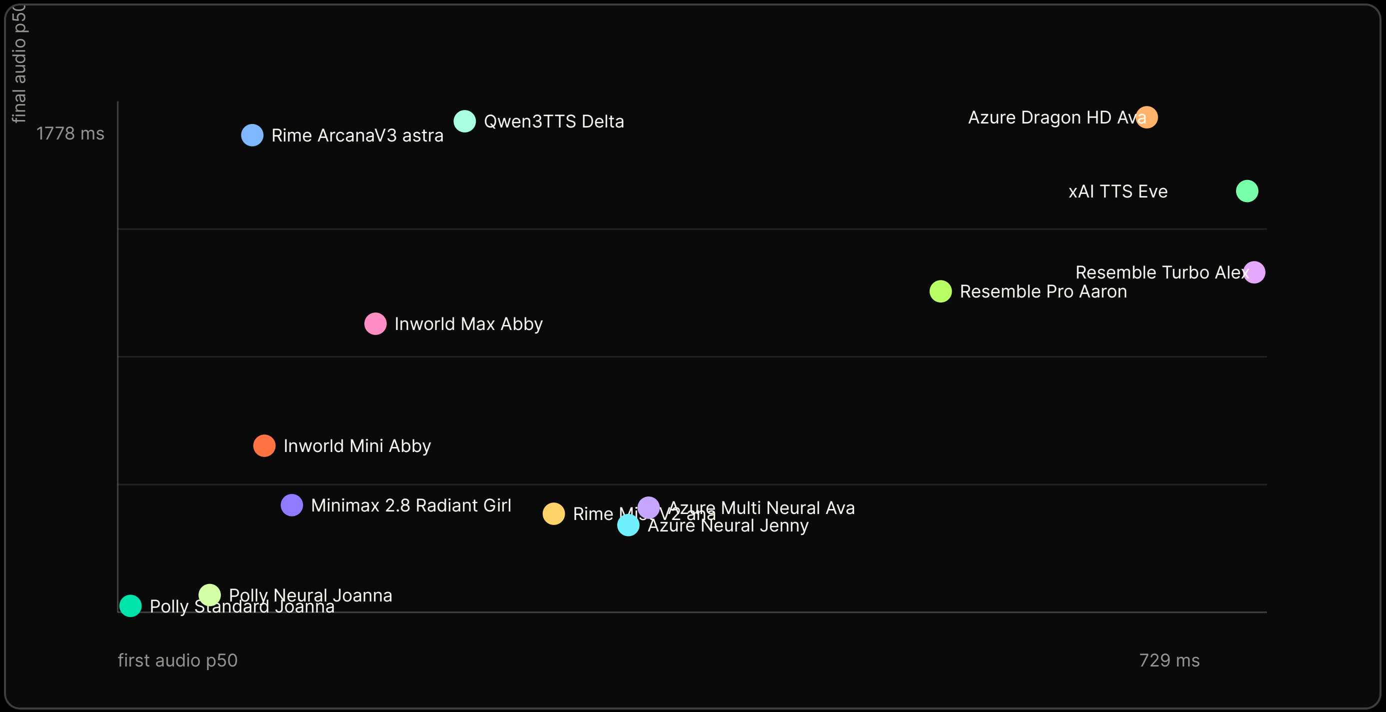
RUN COUNT
6 per voice
Three prompts with two runs per prompt. All measured voices completed 6/6.

FORMAT
mp3
Used for broad compatibility across the selected model families.

ACCESS NOTE
Ultra and Google
Ultra was left out per access caveat. Google voices were not visible in the API.

FIRST AUDIO AND FINAL AUDIO TIMING

Fast first audio is the perceived start delay. Final audio matters when the application waits for the full clip.



WHAT THIS MEANS FOR USE CASES

Use this benchmark as a decision aid, not a universal ranking. TTS choice still depends on voice quality, language, cloning needs, transport, and playback strategy.

LIVE AGENT START

Watch first audio

This is the delay users feel before the assistant starts speaking.

LONGER RESPONSES

Watch final audio

Final audio matters if the app buffers complete audio before playback.

PROVIDER TESTING

Check voice IDs

Model-family names need a concrete voice ID before the benchmark is meaningful.

USE CASE	METRIC TO WATCH	BEST RESULT HERE	READOUT
Live voice agents	First audio p50	Polly Standard, 250 ms	Fastest start in this representative run.
High-quality voice choice	Voice quality plus latency	Rime, Minimax, Qwen3TTS, Azure HD	Latency alone is not enough for the final voice decision.
Custom or cloned voices	Provider family and voice ID	Qwen3TTS, Resemble, Inworld	Use the exact saved voice ID for customer-specific testing.
Coverage claims	Live API availability	Measured set only	Google and Azure Standard were not visible in the API snapshot.

TEST SETUP

Same prompts

Three short English prompts, two runs per prompt.

MEASURED SET

14 voices

Representative voices across accessible model families.

METRIC SOURCE

Runner output

Aggregate p50 and p95 values from the benchmark JSON.

SALES CAVEAT

Do not overclaim

Customer text and target voices should be tested before final choice.

Source: model-coverage-tts-benchmark-results.json

Ultra omitted. Google voices not exposed in live Voices API.